

# SMART INDIA HACKATHON 2026

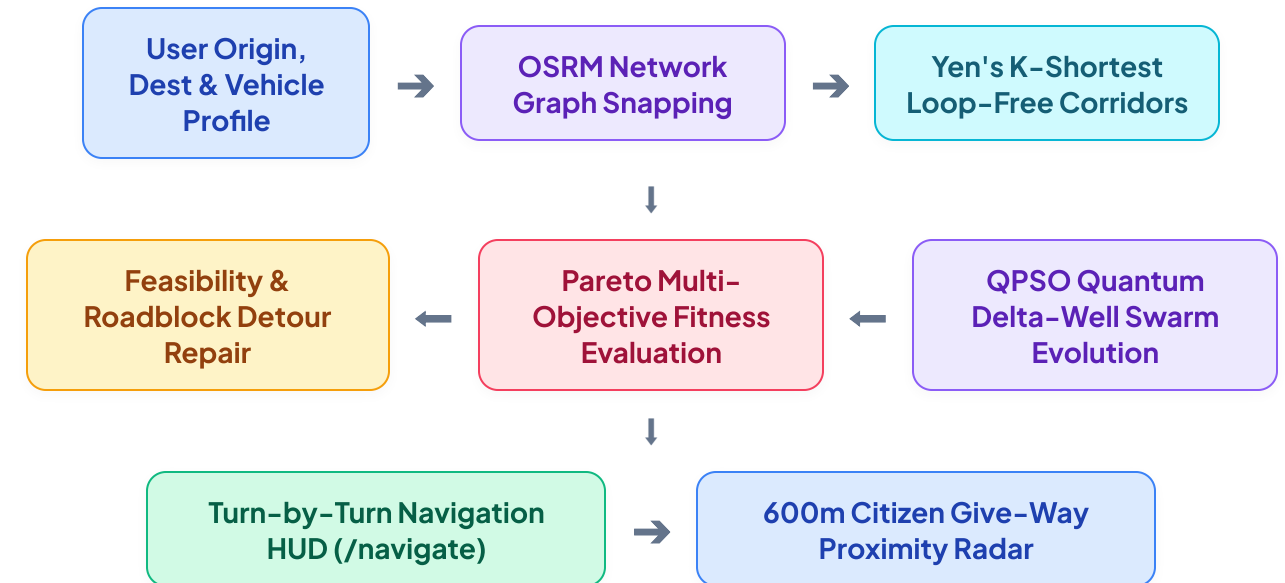
- **Problem Statement ID** – **SIH26137**
- **Problem Statement Title** – Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization
- **Theme** – Transportation & Logistics
- **PS Category** – Software
- **Project / Idea Title** – **MARGSETU ("One Platform. Smarter Routes. Safer Roads.")**
- **Team Name** – **Algo Sapiens**

## Proposed Solution / Approach

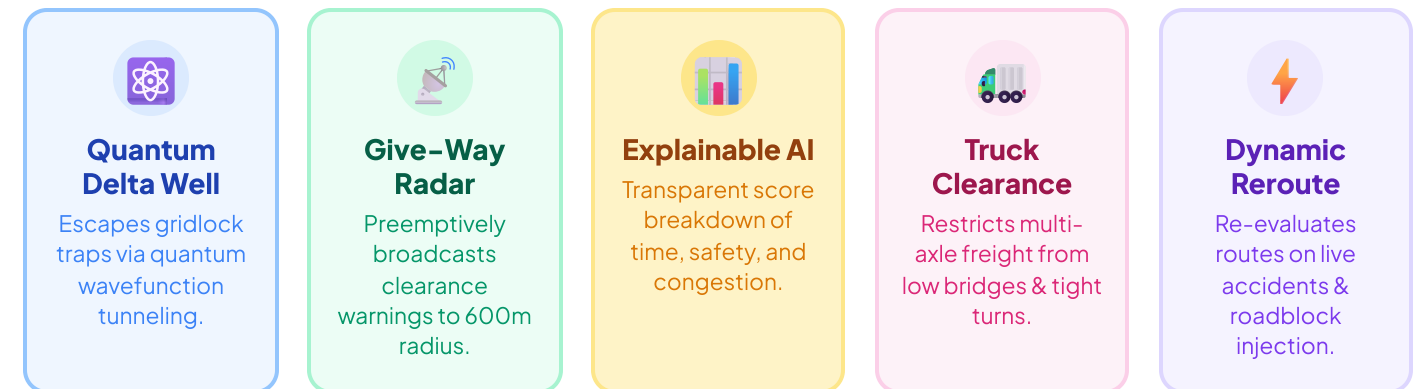
- **Quantum Metaheuristic Search (QPSO Engine):** Eliminates local minima traps and "herd-routing" traffic jams by exploring solution spaces using quantum delta-potential well wavefunctions and Mean Best ( $m_{best}$ ) center-of-mass attraction.
- **5-Factor Pareto Fitness Function:** Simultaneously balances five normalized objectives: **Travel Time (\$T\$)**, **Congestion (\$C\$)**, **Distance (\$D\$)**, **Accident Risk (\$R\$)**, and **Blockage (\$B\$)**.
- **9 Multi-Modal Vehicle Archetypes:** Custom constraints for Cars, Bikes, Transit Buses, Heavy Cargo Trucks, Deliveries, and Emergency Units (Ambulances, Fire Tenders, Police Interceptors).
- **600m Citizen Give-Way Preemption Radar:** Proactively alerts civilian drivers within 600m to clear green corridors before emergency responders arrive.
- **Sub-50ms Dynamic Rerouting:** Continuous WebSocket monitoring prompts faster alternate corridors whenever live road traffic causes  $\geq 1.0$  ext{ min} delay.

• **Explainable AI (XAI):** Transparent percentage breakdowns explain exactly why every route was selected.

## MARGSETU END-TO-END WORKFLOW



## INNOVATION AND UNIQUENESS



## 4-Zone Layered System Architecture

### ZONE 1: CLIENT & EXTERNAL

- Leaflet Map Users
- Citizen Mobile Clients
- Smart City ITS Feeds
- OSM Nominatim API

### ZONE 2: PRESENTATION (UI)

- React 18 + TypeScript
- Vite 5 Bundler
- Leaflet 1.9 Vector HUD
- QPSO Swarm Visualizer

### ZONE 3: BACKEND GATEWAY

- FastAPI (Python 3.10+)
- Uvicorn ASGI Server
- Pydantic v2 Validation
- WebSocket Live Manager

### ZONE 4: QPSO "BRAIN"

- NumPy Vector Swarm
- Delta Wavefunction State
- Dynamic Graph Engine
- Speed Degradation Model

### Quantum Mechanics Formulation:

Particles evolve in a Delta-Potential Well centered at Local Attractor  $p_{i,d} = \phi_d p_{best_{i,d}} + (1 - \phi_d) g_{best_d}$ . State position:  $x_{i,d}(t+1) = p_{i,d} \pm \eta \cdot |m_{best,d} - x_{i,d}(t)| \cdot \ln(1/u)$ . Single parameter  $\eta$  linearly decays, ensuring global exploration early and sharp exploitation upon convergence.

### COMPONENTS / TECHNOLOGY STACK:



## Implementation Process

- Corridor Discovery**  
GPS points snapped to road network; Yen's algorithm & OSRM discover  $K$  diverse, loop-free real road paths.
- Quantum Decision Encoding**  
Continuous particle decision logits  $X \in \mathbb{R}^K$  mapped via Softmax probability archetype selection.
- QPSO Swarm Optimization**  
Calculates Mean Best ( $m_{best}$ ) center of mass; updates particle positions via stochastic quantum wavefunctions.
- Feasibility & Detour Repair**  
Detects vehicle height/weight restrictions and road closures; repairs broken segments with localized Dijkstra detours.
- Multi-Objective Scoring**  
Evaluates normalized fitness  $F = w_1 T + w_2 C + w_3 D + w_4 R + w_5 B$  across time, congestion, and safety.
- HUD Guidance & 600m Radar**  
Dispatches turn-by-turn guidance and streams real-time proximity give-way clearance notifications.

## Feasibility

- **Sub-50ms Execution:** QPSO converges in 35-55 ms, ideal for real-time mobile navigation & dispatch.
- **Zero Quantum Hardware Cost:** Pure software delta-potential simulation running on commodity cloud VMs and browsers.
- **Standard Interoperability:** REST & WebSocket endpoints easily ingest city ITS feeds, IoT sensors, and GPS trackers.

## Viability

- **Universal Market Appeal:** Serves 9 distinct multi-modal vehicle categories across public, commercial, & emergency sectors.
- **Explainable AI (XAI):** Transparent Pareto score breakdowns build immediate trust with commuters and city officials.
- **Emergency Clearance:** 600m give-way radar drastically cuts hospital transit times during the golden hour.

## Practical Implementation

- **Working Prototype Ready:** Fully functioning Map Planner (/map), Navigation HUD (/navigate), & Simulator (/traffic).
- **Ready for Pilot Rollout:** Feasible for city emergency medical and bus fleet trials within 3 to 6 months.
- **Low Bandwidth Architecture:** Lightweight responsive PWA UI functions seamlessly across patchy 4G/5G mobile networks.

## ⚠️ POTENTIAL CHALLENGES AND RISKS

- 01 GPS Telemetry Blindspots:** Urban canyons & underground tunnels causing signal loss.
- 02 Sparse Road Sensors:** Peripheral/rural roads lacking live municipal traffic cameras.
- 03 Peak Concurrency Surges:** Sudden query spikes during rush-hour storms or evacuations.
- 04 Driver Habitual Resistance:** Commuters reluctant to deviate from familiar travel corridors.
- 05 Inter-Agency Data Silos:** Fragmented formats across municipal traffic, police, & EMS.

## 🛡️ STRATEGIES FOR OVERCOMING CHALLENGES

- **01 Dual Client-Side Engine:** In-browser OSRM client continues offline routing seamlessly.
- **02 Synthetic Speed-Flow Model:** Predicts degradation non-linearly with citizen crowdsourcing.
- **03 In-Memory LRU Caching:** Sub-millisecond route caching & Docker horizontal scaling.
- **04 Explainable AI Incentives:** Shows exact minutes saved & fuel preserved to build user trust.
- **05 Open GeoJSON Standards:** Unified open REST/WebSocket interfaces for seamless interop.

“India’s first quantum-inspired intelligent transportation network — smarter routes, faster clearance, safer roads.”

## POTENTIAL IMPACT ON TARGETED AUDIENCE

### Daily Commuters:

**18–25% reduction** in peak urban travel time; prevents herd-congestion and phantom traffic bottlenecks.

### Emergency Responders (Ambulance / Fire):

**30–45% faster response**; active 600m Give-Way radar preemptively clears hospital corridors during golden-hour transit.

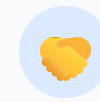
### Commercial Logistics & Cargo Fleets:

**40% fewer grounding incidents** by enforcing bridge height, axle weight, and turn radius compliance.

### Traffic Police & Municipal Smart Cities:

Centralized live congestion injection, dynamic detour routing, and automated green corridor coordination.

## BENEFITS OF THE SOLUTION



### Social & Public Safety

Saves lives by cutting emergency response delays. Transparent Explainable AI route reasoning builds commuter trust and reduces road rage.



### Economic & Commercial

Saves crores in fuel waste and worker productivity. Enhances supply chain throughput by eliminating logistics idle time and detours.



### Environmental Sustainability

Cuts vehicular emissions by **12–15%** via steady-state cruise optimization and eliminating stop-and-go idling (UN SDG 9 & 11).

## Field & Applied Research

- **Urban Traffic Pattern Analysis:** Evaluated congestion bottleneck dynamics, signal queueing, and emergency corridor delays across high-density Indian metropolitan routes.
- **Multi-Objective Algorithmic Benchmarks:** Conducted empirical comparative testing in the MargSetu benchmark lab, demonstrating QPSO's superior Pareto convergence and escape from local minima compared to Dijkstra, A\*, and standard PSO.
- **Real Road Geometric Validation:** Ingested high-resolution OpenStreetMap vector topologies and verified clearance compliance across 9 multi-modal vehicle classifications.

## Academic & Government Sources

- **Sun, J., Feng, B., & Xu, W. (2004):** *"Particle Swarm Optimization with Particles Having Quantum Behavior."* Proceedings of the IEEE Congress on Evolutionary Computation, pp. 325–331. (Mathematical foundation for Delta-Potential Well wavefunction convergence).
- **Yen, J. Y. (1971):** *"Finding the K Shortest Loopless Paths in a Network."* Management Science, Vol. 17, No. 11, pp. 712–716. (Algorithmic foundation for diverse alternative corridor generation in transportation graphs).
- **Ministry of Road Transport and Highways (MoRTH), Govt. of India:** *"Road Accidents in India Annual Report & National Urban Transport Policy (NUTP)."* Official Government traffic accident and mobility benchmarks. <https://morth.nic.in>
- **Bureau of Public Roads (BPR) Urban Planning Formulation:** *"Traffic Assignment Manual."* U.S. Dept. of Commerce (Speed-flow degradation model relating congestion volume to link travel delay).
- **Open Source Routing Machine (OSRM) & OpenStreetMap Foundation:** *"High-Performance Real Road Topological Routing Infrastructure."* <https://project-osrm.org>
- **Smart India Hackathon 2026 Problem Statement SIH26137:** *"Quantum-Inspired Intelligent Traffic Route Optimization in Transportation Systems Using Metaheuristic Optimization."*